

SMART PERSONNEL NEWSPAPER

- 1) Övgü Bejan Taburoğlu – 230201020 (Project Manager)
- 2) Azra Efza Gür – 220201025
- 3) Elif Köken – 220201054
- 4) Fevzi Samed Ünal – 220205032
- 5) Hamit Güneş – 220204042
- 6) İbrahim Türkel – 220201047

1. Introduction

In the digital age, news is produced and distributed at an extremely rapid pace through websites, applications, and social media platforms. Although information is more accessible than ever, this abundance also creates a major challenge for users, especially for busy professionals who need to stay informed in a limited amount of time. They are often exposed to large volumes of repetitive, irrelevant, or unverified content, which makes it difficult to identify truly important news efficiently.

The main problem addressed by the Smart Personnel Newspaper system is the lack of a focused, personalized, and time-efficient news-reading environment. Traditional news platforms generally present the same content to all users without considering their interests, priorities, or reading habits. As a result, users spend unnecessary time searching for meaningful articles and may miss information that is actually relevant to them. This project aims to reduce information overload and improve the quality of the user experience by offering a smarter and more personalized way of consuming news.

1.1. Purpose of the System

The purpose of the Smart Personnel Newspaper system is to design and develop a web-based personalized news platform that collects news from multiple sources and presents it to users in a more relevant, organized, and user-centered manner. The system is intended to help users reach important information quickly instead of wasting time navigating through excessive and unrelated content.

More specifically, the system aims to:

- Deliver personalized news according to user interests,
- Support faster access to significant daily developments,
- Reduce repeated and low-value information,
- Improve the overall efficiency of reading and following the news.

In this way, the project is not only a news display platform, but also a practical and intelligent environment that helps users stay informed more effectively.

1.2. Scope of the System

The Smart Personnel Newspaper system is a software application developed primarily for professionals, students, and other users who prefer to access personalized news through a single digital platform. The scope of the system includes user registration and login, preference selection, personalized news feed generation, article categorization, and content presentation in an accessible interface.

The system will gather news content from external providers such as APIs or RSS-based sources, process this content, and present it according to the user's selected interests and interaction history. It may also support features such as article summaries, daily briefings, and adaptive recommendations based on reading behaviour.

From an administrative perspective, the scope also includes managing categories, monitoring content flow, and maintaining the platform's overall functionality. However, the system does not aim to replace original news publishers or create original journalistic content. Its role is to aggregate, filter, organize, and personalize existing news data for end users.

1.3. Stakeholders

Several stakeholders are involved in or affected by the Smart Personnel Newspaper system:

1) End Users / Readers:

These are the primary stakeholders of the system. They create accounts, select areas of interest, read articles, and interact with the personalized news feed. Their satisfaction and usability needs are central to the success of the system.

2) Administrators:

Administrators are responsible for supervising the platform, managing categories or content-related operations, and ensuring that the system runs properly and efficiently.

3) Developers / Project Team:

The development team is responsible for the analysis, design, implementation, testing, and maintenance of the system. They are key stakeholders because they transform requirements into a functioning software product.

4) News Sources / External Content Providers:

External APIs, RSS feeds, or similar content providers are important stakeholders since the system depends on them for obtaining up-to-date news data.

5) Institution / Organization Management:

The institution or organization management is an important stakeholder because the system can be used to share official announcements, events, and important information with personnel. They benefit from the system by improving internal communication and ensuring that relevant information reaches users effectively.

2. System Overview

2.1. The Overall Functionality of the System

The Smart Personnel Newspaper System is designed to provide users with a personalized digital newspaper by collecting news and information from multiple sources. Instead of following different newspapers, news websites, and digital platforms separately, users can access relevant content through a single system.

The system gathers current news from various sources, organizes the collected content, and presents it according to users' interests and preferences. In this way, each user is provided with a customized newspaper-style interface that highlights the most relevant and useful information.

In addition to personalized news delivery, the system also supports the sharing of institutional events and important announcements. Users can read their personalized newspapers digitally and, when necessary, print them for offline use.

2.2. The Major Components or Modules of the System

The Smart Personnel Newspaper System consists of several major components that work together to provide a personalized and organized news experience. These components are:

- **News Data Collection Module:** Collects news and related content from multiple digital and print-based sources.
- **News Classification and Categorization Module:** Organizes the collected news into predefined categories to make content easier to manage and access.
- **Personal Newspaper Generation Module:** Creates a customized newspaper for each user based on their interests and preferences.
- **Printing Module:** Enables users to print their personalized newspapers when needed.
- **Event and Announcement Module:** Provides users with important institutional events, notifications, and announcements.
- **Search and Filtering Module:** Allows users to search for specific news items and filter content according to their needs.
- **Analytics and Recommendation Module:** Analyzes user interests and interaction patterns to recommend relevant content.
- **Reporting and Administration Module:** Supports administrative tasks such as monitoring the system, managing content, and generating reports.

2.3. The Main Objectives and Capabilities of the System

The main objective of the Smart Personnel Newspaper System is to improve access to information by delivering current and relevant news in a structured and personalized way. The system aims to reduce the difficulty of following multiple information sources and to help users quickly reach the content that matches their interests.

Another important objective of the system is to strengthen internal communication by making institutional events and announcements more visible and accessible. In this way, the system serves not only as a personalized news platform but also as a communication tool for organizational information sharing.

The system is capable of collecting and organizing news, generating personalized digital newspapers, supporting search and filtering operations, recommending relevant content, and enabling both digital viewing and printed output. These capabilities make the system practical, user-focused, and suitable for efficient daily information access.

3. User Types

3.1. The Role of the User In the System

1) End Users / Readers: Their role is to access personalized news content, follow current information, and receive relevant announcements and event notifications.

2) Administrators: Their role is to manage and supervise the system, ensure its proper operation, and control administrative processes.

3) Content Managers / Moderators: Their role is to maintain the quality, relevance, and organization of the news content presented in the system.

3.2. The Main Interactions the User Will Have With the System

1) End Users / Readers: They create accounts, select areas of interest, read personalized news, search and filter articles, view announcements, and print their personalized newspapers when needed.

2) Administrators: They manage users, monitor system activities, control categories and content-related settings, oversee announcements, and handle reporting and administrative tasks.

3) Content Managers / Moderators: They review collected news, organize articles into categories, correct classification issues when necessary, and maintain content quality and consistency.

4. Functional Requirements

1. News Data Collection Module

FR1 - The system shall automatically collect news from daily newspapers, online news portals, and blogs.

FR2 - The system shall integrate with news providers through APIs where available.

FR3 - The system shall support scheduled data fetching on a real-time, hourly, or daily basis.

FR4 - The system shall support collecting news in multiple languages.

FR5 - The system shall track news sources and store citation information.

FR6 - The system shall detect and remove duplicate news content.

2. News Classification & Categorization Module

FR7 - The system shall automatically classify collected news into predefined categories such as Technology, Politics, Sports, and Entertainment.

FR8 - The system shall support multi-label classification for articles belonging to more than one category.

FR9 - The system shall extract keywords and assign tags to news articles.

FR10 - The system shall use AI-based methods for content categorization.

FR11 - The system shall allow administrators to manually reclassify news articles when necessary.

3. Personal Newspaper Generation Module

FR12 - The system shall generate a personalized news feed based on the user's selected interests.

FR13 - The system shall allow users to filter news by date, source, or popularity.

FR14 - The system shall generate a digital newspaper layout for personalized content.

FR15 - The system shall prioritize headlines automatically according to relevance or importance.

FR16 - The system shall include source references and publication dates in the generated newspaper.

FR17 - The system shall generate daily or weekly newspaper editions.

4. Hardcopy Printing Module

FR18 - The system shall export the personalized newspaper in print-ready PDF format.

FR19 - The system shall provide customizable layout templates such as A4, tabloid, and booklet formats.

FR20 - The system shall provide print preview functionality before printing.

FR21 - The system shall support downloading and direct printing of the personalized newspaper.

5. Event & Announcement Module

FR22 - The system shall automatically detect and display important institutional events.

FR23 - The system shall categorize events as academic, social, deadlines, and exams.

FR24 - The system shall send push or email notifications for critical announcements.

FR25 - The system shall provide event reminder functionality.

FR26 - The system shall integrate important events into the personalized newspaper edition.

6. Search & Filtering Module

FR27 - The system shall provide keyword-based search functionality.

FR28 - The system shall provide advanced filtering options based on category, date, source, and popularity.

FR29 - The system shall allow users to bookmark and save articles.

FR30 - The system shall detect and highlight trending news.

7. Analytics & Recommendation Module

FR31 - The system shall track user reading behavior for personalization purposes.

FR32 - The system shall recommend news articles based on previous user interactions.

FR33 - The system shall analyze trending topics.

FR34 - The system shall generate engagement statistics related to user activity.

FR35 - The system shall adapt content ranking according to user behavior and preferences.

8. Reporting & Administration Module

FR36 - The system shall allow administrators to manage news sources and categories.

FR37 - The system shall monitor system performance and logs.

FR38 - The system shall generate usage reports for administrative purposes.

FR39 - The system shall manage user roles and permissions.

FR40 - The system shall provide content moderation tools for administrators.

Additional Requirements

FR41 - The system shall allow users to register and log in securely.

FR42 - The system shall allow users to update their profile information and preferences.

FR43 - The system shall allow users to mark articles as read or unread.

FR44 - The system shall support dark mode and adjustable font size for better readability.

5. Non-Functional Requirements

NFR1 - The system shall employ a decoupled, multi-language architecture to separate core business logic from the data processing pipeline.

NFR2 - The system shall use a centralized relational database, such as PostgreSQL, to store scraped news articles, user permissions, and categorized user profiles.

NFR3 - The personalization algorithm shall be adaptive and shall periodically recalculate its ranking weights to give increasing importance to users' implicit reading behavior over their initial explicit selections.

NFR4 - The system shall perform AI-based classification, sentiment analysis, and summarization asynchronously in order to maintain responsive server performance.

NFR5 - The export engine shall use a unified HTML template pipeline as the single source of truth for generating both email content and static PDF outputs consistently.

NFR6 - The system shall be scalable enough to handle increasing numbers of users, news articles, and requests without significant degradation in performance.

NFR7 - The system shall be compatible with modern web browsers and provide a responsive interface across desktop, tablet, and mobile devices.

6. System Constraints

6.1. Budget Constraints

During the development of the Smart Personnel Newspaper System, it is aimed to keep the project cost at a minimum level. For this reason, the system will be developed using open-source software tools and technologies whenever possible. The use of open-source solutions eliminates licensing costs and supports the economic sustainability of the project.

The project will avoid the use of commercial or paid software licenses unless it is absolutely necessary. Development environments, libraries, frameworks, and other software tools will be selected from free or open-source alternatives. Examples of such tools may include open-source programming languages, databases, and development frameworks.

This approach helps reduce development costs while also making the system easier to maintain and distribute in the future. By relying on open and widely supported technologies, the project can remain flexible and cost-efficient during both development and deployment phases.

6.2. Hardware Constraints

The Smart Personnel Newspaper System is designed to operate without requiring specialized or high-cost hardware. The system will be accessible through standard personal computers, tablets, and smartphones. This ensures that users can access the system easily without the need for additional hardware infrastructure.

Users will be able to access the system using modern web browsers or supported mobile platforms such as Android and iOS devices. Therefore, the system must be compatible with commonly used consumer devices.

For end users, the system will only require a stable internet connection and an up-to-date operating system or web browser. No specialized hardware components will be required for normal system usage.

For development, administration, and testing purposes, a standard computer environment is recommended. The suggested minimum development hardware requirements are as follows:

- Processor: Dual-core processor or higher
- Memory (RAM): At least 8 GB RAM
- Storage: Standard SSD storage unit
- Other: Up-to-date operating system and stable internet connection

These requirements are considered sufficient to support system development, news data processing, testing activities, and system maintenance efficiently.

6.3. Memory Constraints

The Smart Personnel Newspaper System is intended to run on standard consumer devices without requiring excessive memory resources. For this reason, the system should use memory efficiently during normal operation and avoid unnecessary memory consumption. Core functions such as news fetching, content classification, personalized newspaper generation, and search operations should be handled in a way that keeps memory usage at a stable and reasonable level.

Since the system is expected to be accessible on personal computers, tablets, and smartphones, it should remain usable on devices with typical hardware capacities. During development and testing, a standard environment with at least 8 GB RAM is considered sufficient for supporting the main system processes, administrative tasks, and regular maintenance activities.

6.4. CPU Constraints

The Smart Personnel Newspaper System should be developed to operate efficiently on commonly used consumer devices without requiring high processing power. CPU usage should remain at a reasonable level during normal activities such as data collection, filtering, categorization, recommendation generation, and newspaper export.

Particular attention should be given to operations that may require more processing power, such as AI-based classification, summarization, or analysis. These tasks should be managed carefully so that the overall responsiveness of the system is not negatively affected. This approach will help ensure a smooth experience for users while also supporting efficient development, testing, and administrative use.

6.5. Platform Constraints

The Smart Personnel Newspaper System is planned as a platform-independent system that can be used on widely available operating systems. In order to ensure accessibility for different users, the system should be compatible with commonly used environments such as Windows, Linux, and macOS.

Since the main access to the system will be provided through a web-based interface, compatibility with modern web browsers is also important. The system should function properly on browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge. In addition, it should remain accessible on mobile devices through supported browser environments, allowing users to reach the system from desktops, tablets, and smartphones with ease.

6.6. Technology Constraints

The Smart Personnel Newspaper System will be developed using widely supported programming languages, frameworks, and software tools. This is important for ensuring that the system remains maintainable, flexible, and easier to improve in the future. Choosing commonly adopted technologies will also make development and collaboration more practical for the project team.

In line with the project's cost-sensitive approach, open-source and free technologies should be preferred whenever possible. Depending on well-known and actively supported tools reduces technical risk and helps the system remain sustainable during both development and deployment. This also makes future updates, maintenance, and possible expansion more manageable.